

LEARNING MODULE

2

The Time Value of Money in Finance

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	calculate and interpret the present value (PV) of fixed-income and equity instruments based on expected future cash flows
☐	calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given the present value (PV) and cash flows
☐	explain the cash flow additivity principle, its importance for the no-arbitrage condition, and its use in calculating implied forward interest rates, forward exchange rates, and option values

INTRODUCTION

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This learning module applies time value of money principles in valuing financial assets. The first lesson focuses on solving for the present value of expected future cash flows associated with bonds and stocks. In the second lesson, the focus shifts to solving for implied bond and stock returns given current prices. This includes solving for and interpreting implied growth rates associated with given stock prices. The final lesson introduces cash flow additivity, an important principle which ensures that financial asset prices do not allow investors to earn risk-free profits, illustrated with several examples. The material covered in this learning module provides an important foundation for candidates in understanding how financial assets are priced in markets.

LEARNING MODULE OVERVIEW



- The price of a bond is the sum of the present values of the bond's promised coupon payments and its par value. For discount bonds, the price reflects only the present value of the bond's par value.
- The value of a stock should reflect the sum of the present values of the stock's expected future dividends in perpetuity.
- Stock valuation models are classified by the expected growth pattern assumed for future dividends: (1) no growth, (2) constant growth, or (3) changing dividend growth.